

Paraffin-based hybrid rocket engines applications: A review and a market perspective



Alessandro Mazzetti ^{a,*}, Laura Merotto ^b, Giordano Pinarello ^a

^a BlueThink S.p.A., Via Fabio Filzi 5, 20124 Milano, Italy

^b CNR-IENI, Via R. Cozzi 53, 20125 Milano, Italy

ARTICLE INFO

Article history:

Received 22 March 2016

Accepted 29 April 2016

Available online 6 May 2016

Keywords:

Hybrid rocket

Launch vehicles

Micro/nano-satellites

Paraffin

Market analysis

ABSTRACT

Hybrid propulsion technology for aerospace applications has received growing attention in recent years due to its important advantages over competitive solutions. Hybrid rocket engines have a great potential for several aeronautics and aerospace applications because of their safety, reliability, low cost and high performance. As a consequence, this propulsion technology is feasible for a number of innovative missions, including space tourism.

On the other hand, hybrid rocket propulsion's main drawback, *i.e.* the difficulty in reaching high regression rate values using standard fuels, has so far limited the maturity level of this technology. The complex physico-chemical processes involved in hybrid rocket engines combustion are of major importance for engine performance prediction and control. Therefore, further investigation is ongoing in order to achieve a more complete understanding of such phenomena.

It is well known that one of the most promising solutions for overcoming hybrid rocket engines performance limits is the use of liquefying fuels. Such fuels can lead to notably increased solid fuel regression rate due to the so-called “entrainment phenomenon”. Among liquefying fuels, paraffin-based formulations have great potentials as solid fuels due to their low cost, availability (as they can be derived from industrial waste), low environmental impact and high performance.

Despite the vast amount of literature available on this subject, a precise focus on market potential of paraffins for hybrid propulsion aerospace applications is lacking.

In this work a review of hybrid rocket engines state of the art was performed, together with a detailed analysis of the possible applications of such a technology. A market study was carried out in order to define the near-future foreseeable development needs for hybrid technology application to the aforementioned missions. Paraffin-based fuels are taken into account as the most promising segment for market development. The present study is useful for driving future investigation and testing of paraffin-based fuels as solid fuels for hybrid propulsion technology, taking into account the needs of industrial applications of this technology.

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* Corresponding author.

E-mail addresses: alessandro.mazzetti@bluethink.it (A. Mazzetti), merotto@ieni.cnr.it (L. Merotto), giordano.pinarello@bluethink.it (G. Pinarello).

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1. Introduction

Space propulsion technology opens fascinating challenges in the medium-long term that offer new tools for facing old problems.

The first challenge is the chance to achieve a low cost, flexible, safe and reliable mass access to space, thus allowing the exploitation of resources at solar system level, which are becoming increasingly scarce on Earth, such as minerals [1,2] and hydrocarbons [3,4], and exploit new frontiers of energy sources, such as regolith [5] and helium-3 [6]. This offers also a possible solution for other relevant issues, such as overpopulation and soil consumption [7].

The second challenge is the development of hypersonic aeronautical propulsion systems for fast payload transport over inter-continental distances, with reduced environmental impact and reduced costs [8,9].

Hybrid rocket engines (HRE) can serve both purposes due to their unique features, which are briefly summarized in the following part of this Section. Vehicles based on traditional HREs (e.g. using polymer-based fuels) are currently being developed and proven for suborbital flight missions [10,11]. Paraffin-based fuels can be the next step to improve these systems' performance.

The HRE concept is based on the physical separation between fuel and oxidizer. In Fig. 1 the general sketch of a direct HRE is shown, with specific focus on paraffin-based fuels behavior. In this configuration, a solid fuel is paired with a gaseous or liquid oxidizer. This structure results in important advantages of hybrid technology when compared to its competitor propulsion systems, i.e. solid rocket motors, in which the solid propellant grain contains both fuel and oxidizer, and liquid engines, in which both the ingredients are stored in liquid phase [12].

Hybrid rocket propulsion technology has been thoroughly investigated during the last eight decades because it offers several advantages over competitor solutions [13,14], as summarized in Table 1. Nevertheless, its drawbacks have prevented its full development for a long time. In recent years, a renewed interest for this technology was triggered by two main factors: the advances achieved in performance enhancement [15,16], and an increased concern about safety and environmental issues.

Despite the huge amount of investigations carried out during the last decades, hybrid technology improvement efforts still face a number of open problems [17,18], such as combustion instabilities and low regression rate values.

Acceleration fluctuations in hybrid propulsion systems are generally not catastrophic, but they can prevent the use of HRE in

manned missions or in missions involving a payload, which may be damaged by this kind of stress. Therefore, combustion instabilities issues need to be addressed [19] in order to increase the hybrid rockets technology readiness level (TRL).

The main drawback for HRE is the low regression rate values, i.e. the low solid fuel consumption rate, which is due to the HRE flame structure [12,14]. Namely, the physical separation between solid fuel and gaseous/liquid oxidizer flowing over its surface results in a diffusion flame in the boundary layer, rather than a premixed flame like that typical of solid rocket motors. The combustion process is sustained by fuel mass pyrolysis and heat feedback, due to convection and radiation, from flame to fuel surface. A blocked reactive boundary layer is established on the fuel surface. Diffusion flames are governed by fluid-dynamics (mixing time scale) rather than chemical kinetics, thus resulting in lower regression rates. Moreover, the lower combustion efficiency achieved when using paraffin-based fuels results in lower engine performance in terms of both delivered thrust and specific impulse. As a consequence of HRE flame structure, the inter-dependent physical phenomena involved in hybrid rocket propulsion are extremely complex. The flow in the combustion chamber is turbulent, multi-phase, multi-species and chemically reacting, with coupled convective and conductive heat flux, radiative heat flux due to two-phase flow, thermal exchange due to chemical reactions and boundary layer blocking effect due to fuel mass pyrolysis. Because of this complexity, many investigations on HRE behavior (and related strategies to increase regression rate), are experimental [20–23]. Moreover, the different time scales involved in combustion and fluid-dynamics phenomena in HRE result in the need to investigate both lab-scale and full scale HRE, because of the difficulty in determining the scale factor [24–26]. Despite these difficulties in HRE modeling, several numerical studies have been performed in recent years, in particular for traditional HTPB-based fuel formulations [27–33]. Gariani et al. [34] simulated HTPB combustion in a simple 2D slab burner, using a reduced kinetic model for butadiene combustion. The same chemical model for hybrid rocket combustion has been investigated in the work by Mazzetti et al. [29], where improved Partially Stirred Reactor (PaSR) approach for fully turbulent chemistry is suggested as more accurate. Pei et al. [35] proposed a global one-step reaction for butadiene to investigate combustion enhancement when the combustor structure is modified.

Several attempts were made in the last decades to increase the fuel regression rate [36] either using a physical approach (i.e. paraffin-based fuels, swirled flows, multi-port fuel grains) [37–40] and more recently [23], or a chemical approach (i.e. use of high-energetic additives such as nano-metals, metallic hydrides), as in [41–47]. The present work is focused on paraffin-based fuels. Such fuels display a different combustion mechanism when compared to traditional (polymer-based) fuels. When heated, these fuels tend to form a low viscosity and low surface tension melted layer on the fuel surface. The oxidizer flowing over the fuel surface causes an instabilization of the melted layer, resulting in fuel droplets detaching from the surface [15,48]. These droplets are entrained in the oxidizer flow and burn as a spray, with higher combustion surfaces, less energy absorption for phase change and lower blockage effect of the convective heat transfer between flame zone and fuel surface. Therefore, the entrainment effect results in notably higher fuel regression rates when compared to traditional (polymer-based) hybrid fuels [49–51].

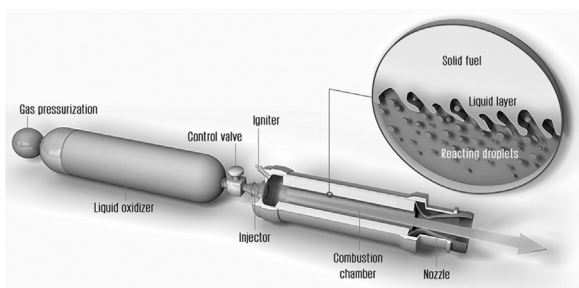


Fig. 1. Sketch of a hybrid rocket engine using a paraffin-based fuel. (Illustration: Emily Cooper).

Table 1
HRE advantages over SRM and LRE technologies.

HRE advantages over solid rocket motors (SRM)	HRE advantages over liquid rocket engines (LRE)
Intrinsic safety during manufacturing, transport, storage, test and launch. Safety is ensured by the physical separation between oxidizer and fuel Reduced cost during development, production and operation. The cost reduction is mainly due to the HRE inherent safety and the structure simplicity Higher performance in terms of specific impulse Oxidizer throttleability, resulting in notable flexibility for mission profiles and better accuracy in orbital injection Robustness with respect to fuel grain crack and debonding, resulting in lower costs of quality control on fuel grain Low environmental impact. Since no perchlorates are used, the environmental impact is reduced with respect to typical launch engines	Increased safety due to minimal fire hazard Better structural mass fraction, due to reduced inert mass of storage/feed and cooling systems Similar performance Architecture simplicity since only one storage/feed system is required and no cooling system is needed for the combustion chamber Higher tolerance for all the feeding system parts, for the oxidizer tank and in general for all the propulsion group

Nevertheless, numerical modeling of paraffin-based fuels is even more challenging than standard fuels modeling. The entrainment effect, typical of paraffin-based fuels [52], requires the modeling of several additional phenomena, such as the instabilization of the thin liquid film on the fuel grain surface, due to oxidizer high-speed flux, and its spraying in the combustion chamber; the transport of fuel droplets, including impacts and coalescence; and the droplets evaporation and combustion behavior, including their contribution to flame front and chemical reactions. Ranuzzi et al. [53] presented a RANS numerical model for paraffin-N₂O hybrid rocket with a simplified global reaction chemistry. In order to further increase hybrid rocket propulsion readiness level, further investigation is needed. In particular, a more complete understanding of the complex phenomena involved in hybrid rocket combustion is required for engine performance prediction and control. Since the regression rate in diffusion combustion depends on the solid fuel gasification conditions, including the thermal energy flux spent for gasification, efforts have been made in order to provide a mathematical modeling of fuel droplet combustion [54–57]. Quasi-equilibrium models and non-equilibrium models provide essentially different results for solid fuel gasification in diffusion combustion, thus confirming the need for experimental validation of the simulations. This considered, it is of significance the renewed interest in visualization experiments for paraffin-based HREs, as in [58]. Other works addressed condensed material surface burning in an oxidizer flow in steady state conditions [59] and in weightlessness [60]. Numerical studies on the liquid film instabilization in HREs have been performed by [61], but a complete modeling of the overall combustion processes in HRE is still not available.

In this framework all the research, development and industrialization activities are of great interest in order to understand the hybrid technology economic perspectives, which are of major importance for market development. In turn, market development perspectives are needed to guide the development of innovative fuels tailored to achieve the goals required for new missions. In this sense, the experience in paraffin-based fuels investigation aimed to obtain low cost, green and reliable fuels is of major importance in the evaluation of perspectives for safe mass access to space and Solar System resource exploitation.

In order to give a complete framework and economic background, the present work takes into account the market analysis performed by various researchers/companies to date, and defines the major development sectors in which a market development is foreseeable. Moreover, an estimation was performed in this work in order to assess the foreseeable advantages, in terms of performance and cost, of replacing standard fuels with paraffin-based fuels for mid-term future innovative hybrid rocket engines based missions.

2. Market perspective

2.1. Hybrid rocket engines for mass access to space: a market analysis

Historically, space flight and orbital injection missions have always been performed by government agencies. Competition and development potential were too limited to allow for a significant enough cost reduction to make the opportunity economically feasible for commercial operators.

A necessary first step for performing new missions in space propulsion, including mass access to space, is to obtain a safe, low-cost and reliable propulsion system. Therefore, it is of major importance to identify a promising propulsion technology to accomplish this task. As already mentioned, hybrid rocket propulsion using paraffin-based fuels can be viewed as a promising solution, and is considered in this work. Moreover, the market potential for this kind of technology needs to be identified, in order to guide future research and development efforts.

Several market studies starting from early 2000 [62], as well as more recent research works [63], identified large growth potential for space tourism as a business concept for profit generation linked to human spaceflight. Federal Aviation Administration 2010 Report [64] forecasts a 1 billion USD turnover by 2030. The economic effort of private companies such as Space Adventures, Space Expedition Corporation (acquired by XCOR Aerospace as of June 2014) and Virgin Galactic show how commercial enterprises start to strongly believe in the success potential of such business. Space Adventures has a perspective of 100 flights at 20 million USD each by 2020. Space Expedition Corporation offers sub-orbital flight tickets (60 km and 100 km altitude) at the price of 100,000 USD each. Virgin Galactic already sold 700 tickets at 200,000 USD market price and has deposits for more than 80 million USD [65]. A relevant paper from Ziliotto [66] summarizes the relevance of the Futron/Zogby space tourism market analysis. In Fig. 2 the passenger demand considering two different market maturation periods (35-year and 45-year) in addition to the baseline of 40 years is shown. By 2021 the market study foresees a passenger number between 10,000 and 20,000 passengers/year, with a baseline scenario of 13,000 passengers/year.

The chart in Fig. 3 shows the 2021 perspective for ticket price reduction. The increase in passenger number, revenue (and possibly competition by several operators) allows for a 75% price reduction between starting and foreseen ticket price in the 2008–2021 timeframe, for a total turnover of nearly 700 million USD [66].

In case of a sufficient and continuous demand, it is possible that the aforementioned companies are able to establish a self-sustained business: this shall result in new competitors entering the market, and therefore to the activation of competition market

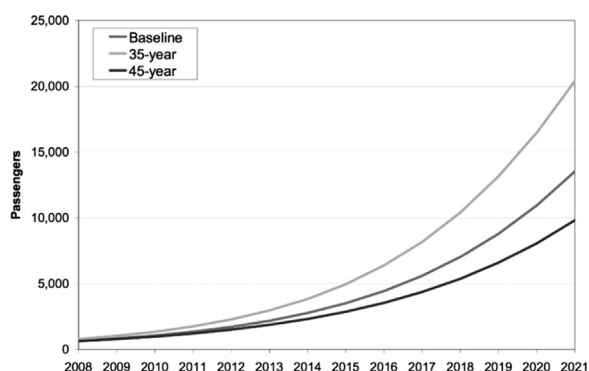


Fig. 2. Futron/Zogby 2006 market research forecast for sub-orbital flight passenger demand, from [67].

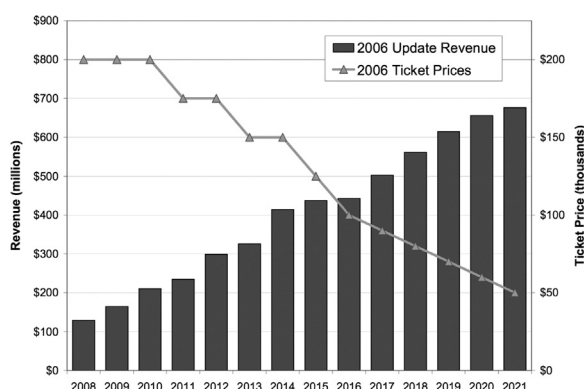


Fig. 3. Futron/Zogby 2006 market research forecast for sub-orbital flight revenue and ticket price [67].

mechanisms leading to increasing reduction in service pricing in order to gain market shares.

In this picture, sub-orbital flight can be seen as the first enabling building block for an incremental development in mass access to space where economic drivers will lead to increasingly higher flight levels: sub-orbital, orbital, lunar, solar system. This escalation trend is not conceptually new as in [68,69], but today is perfectly in line with current operations performed by private companies and governmental institutions. In particular, the first steps (sub-orbital and orbital launches) see an increasing participation by private operators. On the other hand, strong international effort by new space programs (i.e. China, India in addition to EU and US) is being carried out considering moon injection missions [70,71], Mars orbit injection missions [72] and planned manned missions to Asteroid Belt and Mars [73–75].

Only by cost-effective access to space it is possible to start an economically self-sustained industry far from Earth surface, with the potential to exploit industrially the opportunities of extra-terrestrial resources (e.g. raw materials and space-based energy). NewSpace Global [76], a consulting company providing financial analyses related to commercial exploitation of space, identified eight pillars for space market development. These verticals start from existing market segments, such as spacecraft manufacturers and launch vehicle providers, which provide the basic infrastructure. Next two pillars, human spaceflight and microgravity research are as well already in place, mainly under governmental agencies management. Last four verticals represent the more ambitious developments, such as: spaceport infrastructure, in-space services, space resources exploitation and space-based energy. Effort is present in the research community for the analyses related to such developments, such as: spaceport efforts for NewSpace perspective development [77]; commercial exploitation

and in-situ utilization of asteroid resources [78–81];

The new, expanding opportunity related to human spaceflight and mass access to space is of great interest in the short-medium period considering it as a potential market for hydrocarbon-based fuels (including paraffins). These fuels can be a technological and economical enabling factor for access and exploitation of space resources in the mid-long term.

Large growth potential for space tourism suggests the need for an improvement in propulsion technologies. In particular, in order to reduce the service price and to allow a fast development of mass access propulsion systems, a cost reduction is mandatory. For applications related to space propulsion, cost reduction is strongly dependent on the space vehicle design. Therefore, strategies for space propulsion cost reduction rely essentially on two approaches. The first approach is based on the use of lower cost and higher performance rocket engines, like hybrid rocket engines. The second approach involves the use of innovative high performance fuels, such as paraffin-based fuels.

Thermochemical propulsion based on solid or liquid propellants is a fully mature technology: only small performance increase (in particular considering specific impulse) is achievable in short time and with little research effort. On the other hand, hybrid rocket motors show the potential of a game changing technology, able to deliver significant benefits in short time [82,37,15] and. Karabeyoglu, Cantwell and Altman [15,37] pioneered paraffin-based hybrid rocket fuels at Stanford University in 2005: the ultimate target of such studies is to develop hybrid rockets able to perform multiple mission profiles, for both civilian and military purposes.

High performance, low cost fuels have are a smart choice from both the business and the technical points of view. This is evident considering that the players on the market show independently developed vehicles with distinctive characteristics: this results in a different pricing of the market offer (i.e. 100,000 USD vs. 200,000 USD tickets). The cost-reduction driver is evident considering the effort by BlueOrigin llc in the development of a sub-orbital launch vehicle based on liquid oxygen/ liquid natural gas (LOX/LNG) liquid bi-propellant couple. This choice is quite far from the traditional liquid oxygen/kerosene (LOX/RP-1) bi-propellant couple and is a consequence of cost minimization and reliability maximization constraints [83].

This market competition-induced push toward innovation and cost reduction is expected to generate commercial dynamics able to extend the target of the commercial operations in the main direction of extra-atmospheric space.

2.2. Market segments for paraffin-based fuels hybrid rocket engines development

The previous sections describe how commercial and technological development of HRE is of great interest in the framework of mass access to space; in particular, paraffin-based fuels offer an interesting solution for low cost, high performance hybrid rockets.

The experience from these research efforts allows performing trade-off analyses for the identification of the most suitable space/aerospace application for the initial development of paraffin-based hybrid rockets. Four main markets can be identified for such technology, each one with different requirements in terms of performance and cost. Ordering them with increasing barrier of development, they are: launch vehicles upper stages and sub-orbital launch vehicles; microsatellites/nanosatellites orbital injection systems; tactical missiles (including target drones); heavy launch vehicles first stages (i.e. substitution of solid rocket boosters and liquid bipropellant stages).

The next sections discuss in detail these sectors, and provide a description of how paraffin-based systems might result in a

winning market approach.

2.2.1. Sub-orbital launch vehicles/launch vehicles upper stages

In order to allow an effective market entrance of paraffin-based space propulsion systems, the first applications of such technology need to have a low barrier for entering the market, and allow the technology potential full expression. Taking into account this features, the most interesting systems are the smallest ones, such as upper stages. In particular:

- propulsion technologies and systems developed for sub-orbital space flight are easily reengineered for launch vehicles upper stages (and vice-versa);
- upper stage engines are smaller and operate at lower pressures with respect to other systems, such as launch vehicles lower stages/boosters;
- considering Tsiolkovsky equation, the specific impulse is critical in last stages of launch vehicles, where it is not possible to additionally increase the change of velocity (ΔV) by inert mass discard.

Table 2 summarizes the most significant competitor engines/motors currently available.

The comparison between upper stages shows how solid propelled stages are characterized by low specific impulse and by safety issues, while liquid propelled stages have high performance level at the price of several drawbacks, such as high cost, complexity, fuel toxicity and/or sub-optimal fuel usage. An increase in TRL for paraffin-based fueled hybrid systems would allow obtaining high performance avoiding the aforementioned drawbacks.

The interest of hybrid rocket applications in sub-orbital systems is increasing in both commercial and public funded embodiments. Commercial vehicles include Virgin Galactic Space ShipTwo, Copenhagen Suborbitals Tycho Brahe and Whittinghill Aerospace mCLS: these are or will be powered by hybrid rocket motors [102]. An example of public effort involves the HYPROGEO EU-Funded project in the Horizon 2020 framework, related to the development of an hybrid rocket for launch vehicles upper stages, under the leadership of Airbus Defence and Space SAS [103].

The results of the 2012 market report [76] are very interesting to complete the discussion on market potential of suborbital reusable vehicles, in particular considering the perspective of paraffin-based innovative hybrid fuel formulations. These results are shown in Fig. 4. The constrained scenario is based on the presence of large-scale economic factors as demand-reducing drivers. In this case, 213-to 255 seat/cargo equivalents per year are considered for the 10-year forecast. The revenue is estimated in 300 million USD over 10-years. The baseline scenario is based on the current market trends and expectations: seat/cargo equivalents are considered in an increasing trend from 373 in year 1 to 533 in year 10. The growth scenario is based on a significant demand increase: an increment from 1096 seat cargo/equivalents in year 1 to 1592 seat/cargo equivalents in year 10 is considered. Noticeably, the growth scenario suggests a number of passengers which is much lower with respect to previous market studies, such as the 13,000 passenger/year forecast for 2021 from [67].

Additional research work from LeGoff et al. [104] suggests an even more optimistic scenario for suborbital space tourism, based on a worldwide market survey performed by IPSOS Research Company. The foreseen pax per year are 600 for year 1 and between 43,000 and 85,500 on year 16. These estimations are considered to be conservative by the authors, being limited only to the wealthiest (i.e. 20% of Americans with net wealth between 25 million and 50 million dollars; above 50% of Americans with net wealth above 50 million dollars; around 10% and 20% for

Europeans or Japanese of the same wealth level).

2.2.2. Nano/microsatellite launch vehicles

SpaceWorks developed several analyses and forecasts related to micro/nanosatellites launch market. Considering the 2013 nano/microsatellite launch services report [105], it is possible to assume that nano/microsatellites launch is a growing market. Data reported in Fig. 5 show a constant number of launches in the 2010–2013 period, despite the rapid growth of nano/microsatellite industry, with an average value of 80 launches/year. Even when considering launch missions specifically dedicated to nano/microsatellites injection, the total launch number remains constant. This is in contrast with the 54.7% increase of the total number of satellite launches (regardless of their mass class) occurred between 2012 and 2013 (139 in 2012, 215 in 2013).

Two additional considerations need to be taken into account:

- approximately half of 2013 launches was delayed (average delay: 4 months) – 53% of the delays was caused by issues related to the launch vehicle;
- more than half of 2013 launches were not performed at full capacity, i.e. with payload capacity still available.

The historical analysis suggests that the current launch vehicle capacity will not be able to satisfy the future demand, in particular considering the increasing number of requests for micro/nanosatellites. The most interesting strategy to face this issue is to increase the number of launches, in particular considering the need for secondary payload spaces for micro/nanosatellites.

Considering the SpaceWorks 2014 [106] (and updates [107]) and 2015 forecasts for nano/microsatellites [108], the potential market growth is even more evident, with a potential launch requirements between 1900 and 2600 nano/microsatellites in the period 2015–2020. The forecast is shown in Fig. 6.

In order to exploit the increase in market demand already identified in 2013, it will be of great importance to put in place specific strategies, as discussed in the previous sections. After cuts in the budget of government agencies, such as NASA 2013–2014 sequestering, commercial companies (e.g. Orbital Sciences Corporation, SpaceX, BlueOrigin) became active in the market in order to satisfy the launch demand.

In this scenario, hybrid launch vehicles for small payloads can be effectively developed using the knowledge established with sub-orbital applications. This is an important step in the direction of overcoming the historical perspective of nano/microsatellites as secondary payload only. The advantages of such dedicated launch systems are: low cost, flexibility, low environmental impact and orbit/time specificity (which is of great importance for satellites which are generally lacking an on-board propulsion system for apogee/final orbit insertion).

2.2.3. Tactical systems

The market for tactical missile systems has the largest potential volume: considering US only, yearly missile systems production is in the order of magnitude of several hundred thousand units [109]. This can be used a conservative estimation of the worldwide tactical systems market, because several other countries with relevant missile system industry are active (such as UK, France, Germany, Italy, Russia, China and India), but no data is publicly available about their production.

Historically, hybrid rocket motors were not applied successfully to military missile / rocket systems [110]. This is related to the characteristics of hybrid rocket systems based on traditional, polymer-based, fuel formulations:

- tactical missile missions require thrust profiles tailored for the

Table 2

Comparison of significant upper stage engines (liquid) and motors (solid).

	Propellant	Type	Specific impulse (vacuum) [s]	Vector	Notes
Orion 38 [84]	HTPB/AP/Al	Solid	289	Taurus/Pegasus Minotaur C	Low specific impulse (I_s) Low safety Environmental impact Orbital debris Expensive
Orbus 21D [82]	HTPB/AP/Al	Solid	293	Athena	Low I_s Low safety Environmental impact Orbital debris Expensive
Star-37 [84,85]	HTPB/AP/Al	Solid	296	Minotaur C	Low I_s Low safety Environmental impact Orbital debris Expensive
Star-48V [84,85]	HTPB/AP/Al	Solid	285	Antares	Low I_s Low safety Environmental impact Orbital debris Low payload
ANB-3789 [86,87]	AN blend	Solid	290 (estimated)	Super Strypi	Low I_s Low safety Environmental impact Orbital debris Expensive
KM-V2b [88]	HTPB/AP/Al	Solid	302	Epsilon	Low I_s Low safety Environmental impact Orbital debris Low payload
LM-2C 3rd stage [89]	HTPB/Hydrazine	Solid	286	Long March 2	Low I_s Low safety Environmental impact Orbital debris Expensive
PS-4 [90]	MON/MMH	Liquid	308	PSLV	High environmental impact Complex High environmental impact Low safety
Kestrel [91]	LOX/RP-1	Liquid	324	Falcon 1E	Sub-optimal O/F Complex Expensive (Discontinued)
Merlin 1D [91,92]	LOX/RP-1	Liquid	309	Falcon 9	Sub-optimal O/F Complex Expensive
Scorpius Stage 3 [82]	LOX/RP-1	Liquid	323	Falcon Heavy Scorpius	Sub-optimal O/F Complex Expensive
RD-58 [82,93]	LOX/RP-1	Liquid	349–361	Proton-M	Complex (LOX) Expensive
RD-869 [94]	UDMH/ N_2O_4	Liquid	315.5	Vega Dnepr	Toxic fuel Expensive High environmental impact
S5.92 [95]	UDMH/ N_2O_4	Liquid	327	Soyuz-FG/Fregat	Toxic fuel Expensive High environmental impact
S5.98 14D30 [96]	UDMH/ N_2O_4	Liquid	328.6	Zenit Rockot Proton-M	Toxic fuel Expensive High environmental impact
RL 10 [97]	LOX/ H_2	Liquid	451 – 466.5	Atlas V Delta IV	Cryogenic Low density Expensive Complex
Vinci [98]	LOX/ H_2	Liquid	465	Ariane 6	Cryogenic Low density Expensive Complex Under development
BE 3U	LOX/ H_2	Liquid	450 (estimated)	New Shepard	Cryogenic Low density Expensive Complex
CE-7.5 [99]	LOX/ H_2	Liquid	452	GSLV	Cryogenic Low density

Table 2 (continued)

	Propellant	Type	Specific impulse (vacuum) [s]	Vector	Notes
YF-75 [100]	LOX/H ₂	Liquid	440	Long March 3 A	Expensive Complex Cryogenic Low density
LE-5B [101]	LOX/H ₂	Liquid	447	H-IIA	Expensive Complex Cryogenic Low density
RocketMotorTwo RM2	N ₂ O/HTPB	Hybrid	250 (estimated)	SpaceShipTwo	Expensive Complex Low I _s
Advanced hybrid [83]	LOX/Paraffin N ₂ O/Paraffin	Hybrid	340 310 (estimated)	–	Combustion Instabilities Low TRL

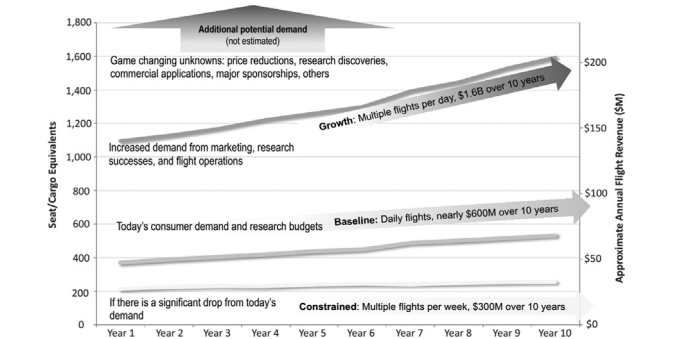


Fig. 4. Constrained, baseline and growth scenarios for suborbital reusable vehicles. Seat/Cargo equivalents are calculated as 1 seat equals 3.5 standard Space Shuttle lockers (16 cu. feet each). From [102].

specific mission, therefore ad hoc geometrical design of the solid propellants fuel grain port are needed;

- in order to achieve high thrust levels, high fuel regression rate is needed. This in turn requires complex, multi-port geometries (i.e. star, wagon-wheel shapes), resulting in production complexity, cost increase and low volumetric efficiency of the system (low fuel mass per unit volume).

The aforementioned requirement of complex multi-port geometries, needed by traditional hybrid rocket fuels to compensate the low regression rate and low thrust, is in conflict with the requirement of a variable geometry fuel grain port tailored for the required mission profile.

Fig. 7 represents relevant mission profiles, their required thrust

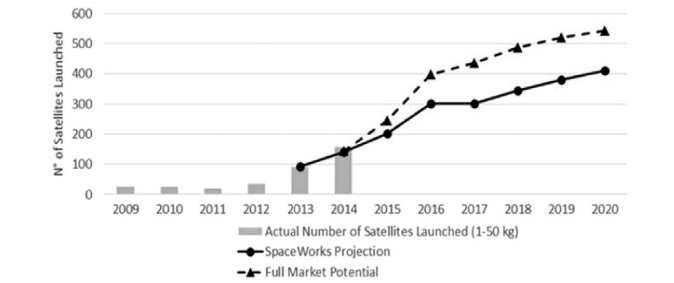


Fig. 6. Actual number of micro/nanosatellites launched and the trend forecast for 2015–2020 period.

profile and the corresponding port geometry.

Paraffin based fuels offer significant benefits in order to overcome the aforementioned limits. They allow conjugating high thrust performance with a simple, single circular central port and thrust tailorability by port variable geometry in order to achieve the desired mission profile. It is worth noticing that paraffin-based fuel grains having non-cylindrical shapes have not been tested so far. Therefore, one can only speculate on the feasibility of realizing a given thrust profile properly shaping the port.

Of course, it is important to ensure that adequate mechanical properties of the fuel grain are maintained for the desired port geometry. This characteristic is viable in order to avoid variable port geometry and reduce fuel grain manufacturing cost. Moreover, hybrid rockets offer the advantage of thrust modulation by oxidizer flux throttling, also investigated for tactical applications (e.g. boost/sustain/boost profiles, such as in [112]). Multiple motor shut-downs and restarts are enabled by a hybrid rocket propulsion

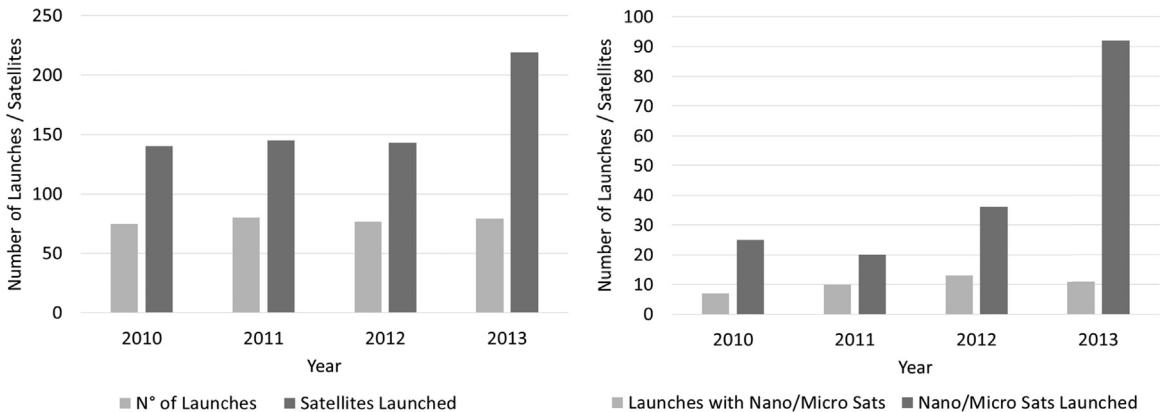


Fig. 5. (Left) Launch and total satellite counts 2010–2013. (Right) Launch and satellite counts, nano/microsatellites only. From [105].

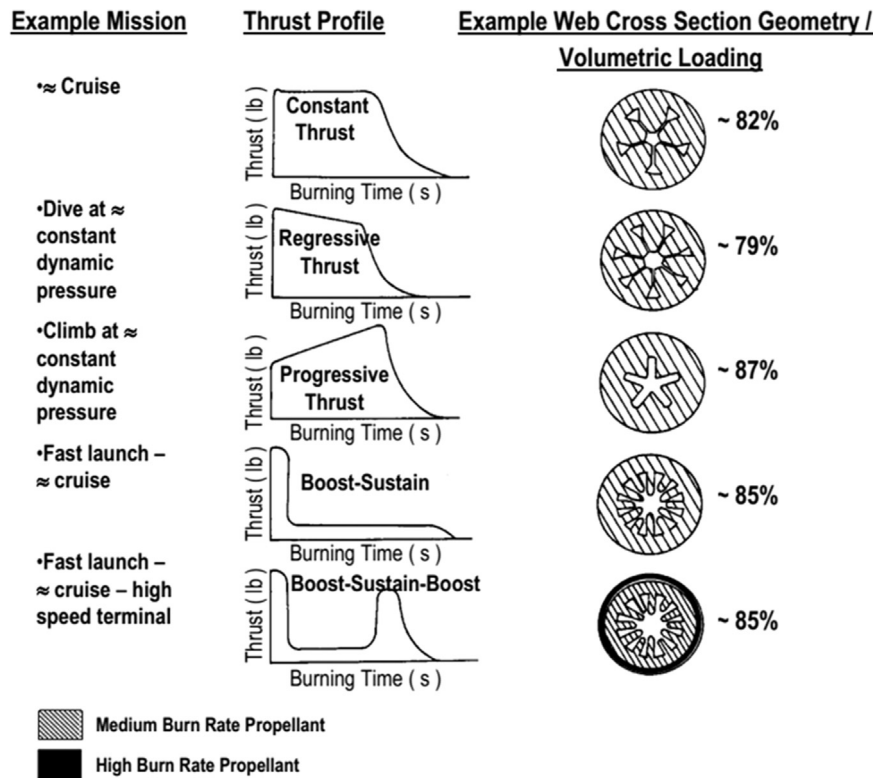


Fig. 7. Mission, thrust profile and corresponding port cross section for solid rockets, from [111].

system. Hybrid rocket motors also show promising improvement over traditional solid propellants in short-range air-to-air missiles scenarios, where the more restrictive constraint related to the use of passive seekers only is in place [113].

In addition, paraffin based hybrid fuels are intrinsically safe (zero-TNT equivalence) in all phases, including storage, transport and handling. This is a definite advantage for the use of such systems in tactical operations scenarios.

2.2.4. Launch vehicles lower stages/boosters

The application of hybrid rocket motors to launch vehicles lower stages and boosters is the most challenging scenario. Lower launch vehicles stages are characterized by very high thrust (magnitude order of several MN), required to reach escape velocity and lift-off of the launch vehicle. This extreme performance level requires a very large system (in the order of 2–4 m diameter, 10–30 m height [114–116,99]). In such geometries, scale-up combustion phenomena occur. Low-scale to large-scale effects involve combustion stability [117], fuel grain mechanical resistance [46] and non-homogeneous fuel consumption issues [118]. The use of traditional HTPB-based hybrid fuels lead to the development of motors up to 1.1 MN thrust, demonstrated successfully by static testing [26,12]. However, multi-port fuel grain was needed to achieve such performance level. The major drawback of this application is the low volumetric efficiency: this lead to prefer solid propellant formulations for the use in lower stages/boosters. This considered, the study from Lengellé et al. [14] concerning the substitution of Ariane 5 solid rocket boosters (two booster configuration, 237 t of propellant each) is of great interest: the use of traditional HTPB-based hybrid solid fuels allows a mass reduction of 23 t, which represents a direct advantage over market competitors. On the other hand, such propellant substitution requires significant modifications, both in geometry and at system level, for the booster. Such modification requires the re-design and re-implementation of all the interfaces between the central launcher

body and the booster, as well as the interfaces between the launch vehicle and the launch platform. If the increase in performance at a lower cost by paraffin-based hybrid fuels is considered, it is possible that such evolution for launch vehicles will take place in the mid-long term.

It is worth considering additional examples from next generation launch vehicles currently under development. For instance, Ariane 6 (EU next generation heavy launch vehicle) is designed to deliver increased payload to GTO orbit using the same propellant technology as the previous Ariane generation: Ariane 6 (A64) is able to deliver 11,000 kg (single satellite payload) while Ariane 5 ECA is able to deliver 9600 kg. Solid propellant mass is 480,000 kg for both Ariane 5 ECA (dual-booster) and Ariane 6 (A64, quad-booster). It is important to consider that without the use of new solid propulsion technology, the payload increase is limited to 1400 kg (less than 15%).

On the other hand, heavy launch vehicles from US commercial operators, such as BlueOrigin, are being developed to deliver their payload to orbit with low cost fuels (LOX/LNG). Paraffin-based HRE might be able to deliver both improvements: an increase in payload given by an increase in fuel performance (higher I_s) without an increase in fuel mass and by using intrinsically low cost fuel.

In fact, it is possible to compare paraffin-based fuel with LNG-based liquid fuel, non-cryogenic liquid fuel (i.e. RP-1) and traditional HTPB/AP/ μ Al solid fuel formulation in terms of performance/cost ratio (i.e. vacuum specific impulse per unit volume

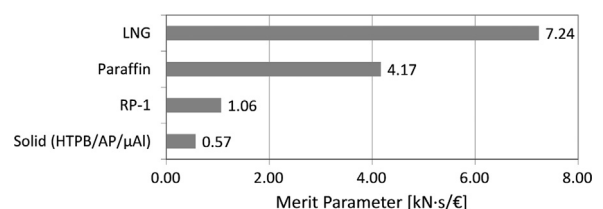


Fig. 8. Comparison of the performance/cost ratio for different fuels.

Table 3
Data used for merit parameter calculation.

	LNG	Paraffin	RP-1	Solid (HTPB/AP/ μ Al)
I_{sp} , vacuum [s]	369	340	358	290
Density [kg/m ³]	820	900	1030	1770
Cost per unit mass [€/kg]	0.5	0.8	3.3	5
Gravity acceleration [m/s ²]	9.81			

over cost per unit volume). This is presented in Fig. 8.

The merit parameter is calculated considering the cost of the fuel only (no additional system-level costs are added) using the following data (Table 3):

I_{sp} , vacuum, gravity acceleration, and density are used to calculate volumetric specific impulse as [kN s/m³]. Density is used to calculate cost per unit volume from cost per unit mass. Considering a merit parameter [kN · s/€], paraffin fuels give a value of 4.17, which is approximately four times RP-1 (1.06) and eight times HTPB/AP/ μ Al solid fuels (0.57). LNG, with a value of 7.24, has the highest merit, but this calculation does not take into account the significant additional system and management costs related to cryogenic liquid fuels.

2.3. Market volume potential for paraffin-based fuels

In order to upgrade hybrid rocket engine technology in view of mass access to space, paraffin has a major role because it allows obtaining low cost, high performance fuels with a low environmental impact. In particular, paraffins can be obtained from industrial waste recycling, thus allowing pollution and cost reduction. Moreover, their combustion mechanism allows reaching higher fuel regression rate values when compared to standard polymeric fuels, thus resulting in higher engine performance. For all these reasons, paraffins have a major role in innovative hybrid rocket engines fuel manufacturing. It is therefore of great interest to investigate which market segments open for the characterization, tailoring, production, and testing of paraffin for hybrid propulsion fuel manufacture.

In order to understand the market potential for the use of paraffin-based rocket fuels, a first-estimation calculation for the potential market volume is needed. It is important to consider that since a significant amount of data is not available to the public or to the research community, only an approximate estimation is possible. The estimation is based on the following assumptions:

- a typical propellant mass value is identified and used for each market segment;
- a 1:1 in mass substitution between currently employed propellants and paraffin-based fuel is considered – this simplification does not consider the specific oxidizer used in each application, and therefore the correct O/F ratio cannot be accounted for. Moreover, the difference in mass due to a

difference in specific impulse (either positive or negative) is not considered;

- the number of produced units/launches per year is based on short term market perspectives for the first market introduction phase of the new technology, and it is used as a conservative approach with respect to growth potential.

The following table summarizes the potential market volumes as estimated.

It is important to consider that the estimation given in Table 4. is valid when the volumes are considered in their magnitude order and not as a precise parameter. This is due to the rapid market evolution possibility for the short term. Therefore, an estimation between the tens and the hundred thousand tons per year of paraffin-based fuels seems in line with currently available data, under conservative estimations.

3. Concluding remarks

Hybrid rocket engines (HRE) are nowadays considered as a promising propulsive solution for important innovative missions, such as mass access to space. This is due to their inherent advantages over traditional aerospace propulsion systems, such as solid rocket motors and liquid rocket engines. On the other hand, hybrid rocket propulsion main drawback, i.e. the low regression rate values using standard fuels, has so far limited hybrid technology maturity level. Many international investigations are currently ongoing in order to overcome such limits using different approaches. One of the more promising methods involves the use of liquefying fuels, such as paraffins, which ensure good performance at low cost, availability (also as industrial byproduct), low environmental impact.

Despite the notable effort for increasing the HRE technology readiness level, no detailed review of the market potential of this kind of technology is available.

In this work a review of hybrid rocket engines state of the art was performed, together with a detailed analysis of the possible applications of such a technology. The four main market segments identified for HRE technology are: launch vehicles upper stages and sub-orbital launch vehicles; microsatellites/nanosatellites orbital injection systems; tactical missiles (including target drones); heavy launch vehicles first stages (i.e. substitution of solid rocket boosters and liquid bipropellant stages).

A market study was carried out in order to define the near future foreseeable development needs for hybrid technology application to the aforementioned missions. Paraffin-based fuels are taken into account as the most promising segment for such market development.

The most relevant conclusions are:

- considering most recent suborbital flight market studies, within a few years an increase up to 20,000 passengers and a 75% per-ticket cost reduction, both over a 13 years timeframe, and a

Table 4
Potential market volume for paraffin-based fuels, first approximation estimation.

Market	Fuel mass range [t]	Launch number potential [launches/year]	Unit produced per year [dimensionless]	Potential market volume [t/year]
Launch vehicles upper stages	1–60	100–1000	–	100–60,000
Sub-orbital flight	3–10	100–1000	–	300–10,000
Micro-satellite launch Vehicles	1–60	> 100	–	100–6000
Tactical systems	0.01–50	–	10 ² to 10 ⁶ globally, depending on type	up to 50,000,000
Launch vehicles lower Stages/Boosters	60–600	100–1000	–	6000–600,000
TOTAL				6500–50,676,000

turnover of about 700 million USD are foreseen;

- in the same years 1900–2600 nano/microsatellites will be operating, all of them requiring market-competitive launch solutions;
- considering the market segments analyzed in this work, replacing traditional propulsive solutions with HRE would allow a notable cost reduction in the mid-term future (35–45 years);
- considering the same market segments, replacing traditional HRE fuels with paraffin-based fuels would allow a relevant cost reduction: paraffin-based fuels have a performance/cost ratio which is 4 times the performance/cost ratio of traditional fuels;
- the full potential market volume for paraffin-based fuels is up to over 50 M t/year, with a minimum of 6500 t/year if no tactical applications are considered.

This study is useful for driving the oncoming investigation and testing of paraffin-based fuels as solid fuels for hybrid propulsion technology, taking into account the needs of industrial applications of such a technology.

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